

1. Which of the following streptococci groups are predominantly found in the oral cavity and are commonly referred to as 'oral' streptococci?
 - a. *S. pyogenes*
 - b. *S. agalactiae*
 - c. *S. anginosus*
 - d. *S. viridans*
2. What is the primary role of 'oral' streptococci, particularly *S. mutans*, in the development of dental caries?
 - a. They produce enzymes that strengthen tooth enamel
 - b. They produce antibiotics that protect teeth from decay
 - c. They produce acid that contributes to the demineralization of teeth
 - d. They help maintain gum tissue health
3. *Clostridium difficile* is usually managed with
 - a. vancomycin ✓
 - b. penicillin v
 - c. clarithromycin
 - d. none of these
4. What is the role of streptococcal pyrogenic exotoxins (erythrogenic toxins) produced by *Streptococcus pyogenes*?
 - a. Facilitate the formation of fibrin mesh
 - b. Break down hyaluronic acid in connective tissue
 - c. Facilitates the release of cytokines, potentially leading to shock
 - d. Lysing erythrocytes and leukocytes
5. Consumption of meat contaminated with *Bacillus anthracis* will lead to
 - a. haemorrhagic diarrhoea
 - b. fever
 - c. malaise
 - d. pruritic
6. Which of the following statements accurately describes the epidemiology of enterococci?
 - a. Predominantly found in the respiratory tract
 - b. Commonly associated with skin infections
 - c. Commensals of the gastrointestinal and vaginal tract
 - d. Found in the upper respiratory tract
7. What is the primary cause of anaemia in malaria infections?
 - a. Liver rupture
 - b. Gametocytes migration
 - c. Schizonts toxins
 - d. Erythrocyte haemolysis
8. What is a notable feature of *Enterococcus* species in terms of growth?
 - a. They require strict anaerobic conditions to grow
 - b. They are motile and exhibit swarming behaviour
 - c. They can grow over a wide temperature range ✓

- d. They form spores as part of their life cycle
9. Symptoms of *Corynebacterium diphtheria* infection include the following, except.
- a. Bullneck (oedema of the neck)
 - ☒ b. Diarrhoea
 - c. Fever
 - d. Fatigue
10. Which of the following antibiotics is a cell inhibitor?
- ☒ a. Penicillin
 - b. Metronidazole
 - c. Vancomycin
 - d. Clarithromycin
11. What is the primary mechanism by which the neurotoxin produced by *C. tetani* leads to the characteristic motor spasms seen in tetanus?
- a. Inhibiting blood clotting factors
 - ☒ b. Blocking neurotransmitter release
 - c. Disrupting cell membrane integrity
 - d. Stimulating excessive muscle contraction
12. Which of the following is the primary mode of action of the neurotoxin produced by *C. botulinum*?
- a. Stimulating excessive muscle contraction
 - ☒ b. Enhancing neurotransmitter release
 - c. Blocking the release of acetylcholine
 - d. Inhibiting blood clotting factors
13. The two medically important *Neisseria* spp are
- ☒ a. *N. gonorrhoeae* and *N. meningitidis*
 - b. *N. meningitidis* and *N. saprophyticus*
 - c. *N. gonorrhoeae* and *N. saprophyticus*
 - d. *N. saprophyticus* and *N. coli*
14. Which bacterial species possesses a protein capsule that prevents phagocytosis and produces a composed of three proteins?
- a. *Bacillus cereus*
 - b. *Clostridium tetani*
 - ☒ c. *Bacillus anthracis*
 - d. *Actinomyces israelii*
15. Which of the following *Neisseria* species is commonly associated with meningitis?
- a. *N. gonorrhoeae*
 - ☒ b. *N. meningitidis*
 - c. *N. lactamica*
 - d. *N. sicca*
16. Enterobacteriaceae are normal commensals of the
- a. nose
 - b. skin
 - ☒ c. intestine
 - d. lungs
17. What is the main role of cell-surface pili in *N. gonorrhoeae* infections?
- a. Aiding in serum lysis
 - b. Enhancing carbohydrate fermentation

- ☒ c. Facilitating adherence to mucosal surfaces
 d. Providing resistance to antibiotics
18. Which of the following amoeba species causes amoebic dysentery?
- a. *Amoeba proteus*
☒ b. *Entamoeba histolytica*
 c. *Amoeba limicola*
 d. *Amoeba vespertilio*
19. Which *Plasmodium* species is responsible for a significant majority of cases in tropical areas?
- a. *P. vivax*
 b. *P. ovale*
☒ c. *P. falciparum*
 d. *P. malariae*
20. What is the primary method of transmission for *Neisseria gonorrhoeae*?
- a. Vector-borne transmission
 b. Faeco-oral transmission
 c. Droplet transmission
☒ d. Sexual contact
21. What is the primary method of transmission for *Salmonella* infections?
- a. Vector-borne transmission
☒ b. Faeco-oral transmission
 c. Droplet transmission
 d. Sexual contact
22. All the following treponema species cause diseases in humans. **except**?
- ☒ a. *T. carateum*
 b. *T. pallidum*
 c. *T. pertenue*
 d. *T. paraluisccuniculi*
23. How are *Salmonella* serotypes typically classified?
- a. By their shape and arrangement
 b. By their motility pattern
 c. By their biochemical characteristics
☒ d. By their O and H antigens
24. Microbiology involves the study of ...
- a. molecules that are visible to human eyes
 b. animals and their family
☒ c. organisms that are not visible to naked eyes
 d. microorganisms using a microscope
25. Who is known as the father of Microbiology?
- a. Edwin John Butler
 b. Ferdinand Cohn
 c. Robert Koch
☒ d. Antoni van Leeuwenhoek
26. Which microorganism(s) perform photosynthesis by utilizing light?
- a. Cyanobacteria, Fungi, and Viruses
 b. Viruses
☒ c. Cyanobacteria

d. Fungi

27. Which part of the compound microscope helps in gathering and focusing light rays on the specimen to be viewed?
- ☒ a. Condenser lens
 - b. Magnifying lens
 - c. Objective lens
 - d. Eyepiece lens
28. Which of the following are produced by microorganisms?
- a. Alcoholic beverages
 - b. Fermented dairy products
 - c. Breads
 - ☒ d. All of the mentioned
29. What is the approximate size of a bacterial cell?
- a. 1 mm in diameter
 - ☒ b. 0.5 to 1.0 micrometer in diameter
 - c. 2 mm in diameter
 - d. 2 micrometers in diameter
30. Which among the following is a "Spirochetes"?
- a. *Streptomyces sp.*
 - ☒ b. *Treponema pallidum*
 - c. *Streptococcus viridans*
 - d. *Corynebacterium diphtheriae*
31. Bacteria having clusters of flagella at both poles of cells are known as?
- ☒ a. Amphitrichous
 - b. Monotrichous
 - c. Peritrichous
 - d. Lophotrichous
32. What is the mechanism of action of beta-lactam antibiotics (e.g., penicillin)?
- a. Inhibition of protein synthesis
 - b. Disruption of cell membrane
 - c. Inhibition of DNA replication
 - ☒ d. Inhibition of cell wall synthesis
33. What is the function of restriction enzymes in bacteria?
- a. Repair damaged DNA
 - b. Facilitate transcription
 - ☒ c. Degrade foreign DNA
 - d. Synthesize RNA
34. Which of the following is a bacterial toxin that causes muscle paralysis by blocking acetylcholine release at neuromuscular junctions?
- ☒ a. Botulinum toxin
 - b. Tetanus toxin
 - c. Cholera toxin
 - d. Diphtheria toxin
35. What is the significance of the polymerase chain reaction (PCR) in molecular biology?
- ☒ a. Amplification of DNA
 - b. Protein synthesis

- c. Repair of damaged RNA
 - d. Inhibition of viral replication
36. A 2-day-old premature baby boy born to a 22-year-old woman developed pneumonia. The mother developed symptoms of urinary tract infection after delivery. A blood sample from the newborn showed the growth of beta-hemolytic colonies within 24 hours. Which group B streptococcus could be the possible pathogen causing pneumonia?
- a. *Streptococcus pyogenes*
 - ☒ b. *Streptococcus agalactiae*
 - c. *Enterococcus faecalis*
 - d. *Streptococcus bovis*
37. Which bacterium is commonly isolated as the pathogen in pharyngitis (sore throat)?
- a. *Streptococcus pneumoniae*
 - b. *Staphylococcus aureus*
 - ☒ c. *Streptococcus pyogenes*
 - d. *Mycoplasma pneumoniae*
38. A 4-year-old boy presents with an erythematous rash on his face and arms, accompanied by swollen lymph nodes around the infection site. The pathogen identified is a gram-positive coccus. What could be the possible diagnosis?
- a. Glomerulonephritis
 - ☒ b. Erysipelas
 - c. Impetigo
 - d. Cellulitis
39. Which characteristic does not apply to *Streptococcus pneumoniae*?
- a. Colonies are diplococcus and encapsulated
 - b. Alpha hemolytic (greenish zone) colonies appear on blood agar
 - c. Found as normal flora in healthy individuals
 - ☒ d. Grows well in a high concentration of glucose media
40. Which bacterium is a common pathogen in otitis media (middle ear infection)?
- ☒ a. *Streptococcus pneumoniae*
 - b. *Streptococcus agalactiae*
 - c. *Streptococcus pyogenes*
 - d. *Streptococcus intermedius*
41. The process of staining bacteria to differentiate them under a microscope is called
- a. Culturing
 - ☒ b. Gram staining
 - c. Replication
 - d. Inoculation
42. What does it mean for a bacterium to be "motile"? It can/is ...
- a. reproduce asexually
 - b. survive in harsh environments
 - ☒ c. move on its own
 - d. pathogenic to humans
43. How does antibiotic resistance develop in bacteria?
- a. Through exposure to antibiotics
 - b. By inheriting resistant genes
 - ☒ c. Both a and b

- d. Neither a nor b.
44. Which of the following is a Gram-negative bacterium?
- a. *Staphylococcus aureus*
 - ☒ b. *Escherichia coli*
 - c. *Streptococcus pyogenes*
 - d. *Clostridium tetani*
45. What is the primary function of bacterial endospores?
- a. Reproduction
 - b. Nutrient storage
 - ☒ c. Survival under adverse conditions
 - d. Antibiotic production
46. Which of the following is a fungal infection affecting the skin, hair, or nails?
- a. Malaria
 - b. Candidiasis
 - ☒ c. Tinea
 - d. Tuberculosis
47. Which type of immunity is acquired through vaccination or exposure to pathogens?
- a. Innate immunity
 - b. Passive immunity
 - ☒ c. Active immunity
 - d. Adaptive immunity
48. What is the causative agent of tuberculosis?
- a. *Streptococcus pneumoniae*
 - ☒ b. *Mycobacterium tuberculosis*
 - c. *Haemophilus influenzae*
 - d. *Legionella pneumophila*
49. Which of the following is a sexually transmitted infection caused by a spirochete bacterium?
- a. Chlamydia
 - b. Gonorrhea
 - ☒ c. Syphilis
 - d. Herpes
50. What is the function of ribosomes in a bacterial cell?
- a. Synthesizing DNA
 - b. Producing ATP
 - ☒ c. Protein synthesis
 - d. Cell division
51. Which of the following is a protozoan parasite causing malaria?
- ☒ a. *Plasmodium vivax*
 - b. *Trypanosoma cruzi*
 - c. *Giardia lamblia*
 - d. *Entamoeba histolytica*
52. What is the primary function of the bacterial capsule?
- ☒ a. Attachment to surfaces
 - b. Nutrient absorption
 - c. Oxygen exchange
 - d. DNA replication

53. Which of the following is a Gram-positive bacterium responsible for causing food poisoning?
- a. Salmonella enterica
 - ☒ b. Staphylococcus aureus
 - c. Escherichia coli
 - d. Campylobacter jejuni
54. Which hosts are involved in the life cycle of Plasmodium parasites?
- a. Humans only
 - b. Mosquitoes only
 - ☒ c. Humans and mosquitoes
 - d. Birds and mosquitoes
55. Who was the first scientist to observe and describe microorganisms accurately?
- a. Robert Hooke
 - ☒ b. Anthon Van Leeuwenhoek
 - c. Louis Pasteur
 - d. Robert Koch
56. All the following form part of Koch's postulate except.
- ☒ a. The microorganism must be present in some cases of the disease but absent from healthy organisms.
 - b. The suspected microorganism must be isolated and grown in a pure culture.
 - c. The same disease must result when the isolated microorganism is inoculated into a healthy host.
 - d. The same microorganism must be isolated again from the diseased host.
57. In the history of medicine, Joseph Lister is well known for.....
- a. tyndallization
 - b. vaccination
 - c. culturing of bacteria
 - ☒ d. antiseptic Surgery
58. What is the term used to describe the process by which bacteria lose their ability to cause disease after incubation for long intervals between transfers?
- a. Vaccination
 - b. Incubation
 - ☒ c. Attenuation
 - d. Inactivation
59. Who is credited with the discovery of vaccination and the term "vaccine"?
- a. Louis Pasteur
 - b. Joseph Meister
 - ☒ c. Edward Jenner
 - d. Shibasaburo Kitasato
60. How did Pasteur prepare the rabies vaccine?
- a. By growing the bacteria at a high temperature
 - b. By injecting inactivated toxin into rabbits
 - c. By incubating the bacteria with potassium bichromate
 - ☒ d. By growing the bacteria in a rabbit host and then drying their brains and spinal cords
61. Which family do herpesviruses belong to?
- ☒ a. Herpesviridae
 - b. Flaviviridae
 - c. Adenoviridae

- d. Papillomaviridae
62. What term is used to describe the process in which blood leukocytes engulf disease-causing bacteria?
- ☒ a. Phagocytosis
 - b. Vaccination
 - c. Attenuation
 - d. Incubation
63. What is the key difference between Gram-positive and Gram-negative bacteria?
- a. Their ability to form spores
 - b. Their resistance to antibiotics
 - ☒ c. The quantity of peptidoglycan in their cell wall
 - d. Their preference in certain environments
64. What is the primary structural difference between the cytoplasmic membranes of bacterial cells and mammalian (eukaryotic) cells?
- ☒ a. Presence of sterols
 - b. Composition of phospholipids
 - c. Number of protein layers
 - d. Thickness of the membrane
65. Mesosomes in bacterial cells are formed by:
- ☒ a. Folding of the cytoplasmic membrane
 - b. Fusion of ribosomes
 - c. Extracellular enzymes
 - d. Mitochondrial activity
66. Which component is a major contributor to the rigidity of the cell wall in Gram-positive bacteria?
- ☒ a. Peptidoglycan
 - b. Teichoic acids
 - c. Lipopolysaccharides
 - d. Phospholipids
67. The primary function of the outer membrane in Gram-negative bacteria is to:
- a. Store energy
 - b. Facilitate cell division
 - c. Regulate nutrient uptake
 - ☒ d. Protect against harmful substances
68. What is the purpose of uncoating in the viral replication cycle?
- a. Attachment to host cells
 - b. Release of viral particles
 - c. Production of viral mRNA
 - ☒ d. Removal of viral protein coat
69. Which of the following is a gram-positive bacterium?
- ☒ a. Staphylococcus
 - b. Streptococcus
 - c. Neisseria
 - d. Bacillus
70. What is the primary function of capsules in bacterial cells?
- a. Facilitate cell division
 - b. Aid in respiration
 - ☒ c. Protect against antibiotics

80. Which is the main component of the virus core, associated with core proteins?
- Lipid bilayer
 - Envelope
 - Nucleic acids
 - Capsid
81. In the growth curve of bacteria, during which phase does the growth rate slow down as nutrients become exhausted and waste products accumulate?
- Lag phase
 - Log phase
 - Stationary phase
 - Decline phase
82. Which of the following is a method used to detect virus replication in cell cultures?
- Polymerase Chain Reaction (PCR)
 - Immunofluorescence
 - Hemagglutination
 - Serological testing
83. Which polysaccharide extracted from seaweed is used to solidify liquid growth media for bacterial culture?
- Cellulose
 - Starch
 - Agar
 - Chitin
84. Which type of media is designed to suppress the growth of certain bacteria while allowing the growth of others?
- Enriched media
 - Simple media
 - Selective media
 - Indicator media
85. How does MacConkey agar aid in the recognition of specific bacterial pathogens?
- It provides additional nutrients for fastidious bacteria
 - It measures bacterial growth rate
 - It detects acid production through pH indicators
 - It enhances bacterial waste product production
86. The binomial system for bacterial naming consists of:
- Genus name and species name
 - Genus name and common name
 - Kingdom name and family name
 - Family name and order name
87. Which of the following is **NOT** one of the characteristics used in the official taxonomy of viruses?
- Nature of the nucleic acid in the virion
 - Symmetry of the capsid
 - Presence or absence of an outer lipid envelope
 - Mode of transmission
88. Which of the following scientists is noted for carrying out experiments with decaying meat to discredit the spontaneous generation theory?
- Francesco Stelluti

- ☒ b. Francesco Redi
 c. Francastoro Girolamo
 d. Aristotle
89. All the following are gram-positive spherical bacteria, except.
- S. aureus*
 - S. epidermidis*
 - E. faecalis*
 - ☒ d. *E. coli*
90. Which of the following bacteria will give positive results for a coagulase test?
- S. epidermidis*
 - S. saprophyticus*
 - ☒ c. *S. aureus*
 - S. hominis*
91. Which enzyme is produced by *S. aureus* coagulates plasma, resulting in fibrin deposition and interference with phagocytosis?
- Lipase
 - Haemolysin
 - ☒ c. Coagulase
 - Hyaluronidase
92. Which of these organisms is a cestode?
- Fasciola hepatica*
 - ☒ b. Tape worm
 - Paragonimus westermani*
 - Guinea worm
93. What type of toxin produced by *S. aureus* causes food poisoning and is resistant to gastrointestinal enzymes and heat?
- Leukocidin
 - Haemolysin
 - ☒ c. Enterotoxin
 - Exfoliative toxin
94. What is a defining characteristic of herpesviruses after infection?
- Rapid clearance from the body
 - ☒ b. Establishment of lifelong latent infection
 - Immediate production of symptoms
 - Inhibition of nucleic acid metabolism
95. All the following are nematodes, except.
- Ascaris lumbricoides*
 - Toxocara canis*
 - Enterobium vermicularis*
 - ☒ d. *Taenia solium*
96. Which symptom of malaria is related to the release of toxins when schizonts burst?
- Anaemia
 - Confusion
 - ☒ c. Fever/sweats
 - Coma

Staphylococcus aureus is frequently associated with infections of implanted devices such as cerebrospinal shunts and hip prostheses due to its ability to:

- a. Produce exfoliative toxins
- b. Form capsules that protect against antibiotics
- ☒ c. Attach firmly onto artificial surfaces and produce a slime layer
- d. Rapidly replicate in the bloodstream

98. All the following are spore-forming bacteria, except.

- a. *Clostridium perfringens*
- b. *Clostridium difficile*
- c. *Clostridium tetani*
- ☒ d. *Escherichia coli*

99. *Streptococcus pneumoniae* partially lyse erythrocytes, therefore what type of haemolysis will be observed on a blood agar?

- ☒ a. a-haemolysis
- b. b-haemolysis
- c. g-haemolysis
- d. d-haemolysis

100. Which of the following is not a viral disease?

- a. Mumps
- b. Measles
- c. Hepatitis
- ☒ d. Pertussis

- d. Aid attachment to host cells and protect against host defenses
71. Extracellular slime layers produced by bacteria are composed predominantly of
- Lipids
 - Proteins
 - ☒ Complex polysaccharides
 - Nucleic acids
72. Which condition can *Campylobacter enteritis* mimic?
- Typhoid fever
 - ☒ Acute appendicitis
 - Malaria
 - Tuberculosis
73. Fimbriae are important for bacterial:
- ☒ adherence to surfaces
 - genetic material transfer
 - formation of biofilms
 - synthesis of flagella
74. Capsid is associated with.....
- ☒ Viruses
 - Bacteria
 - Metazoan
 - fungi
75. HIV-1 belong to the family of
- reoviridae
 - ☒ retroviridae
 - parvoviridae
 - poxviridae
76. Which term describes a complete infectious virus particle?
- Nucleotide
 - Capsomere
 - Envelope
 - ☒ Virion
77. Herpes simplex virus can cause the following infections, except
- conjunctivitis
 - cold sores
 - pneumonia
 - ☒ fever
78. Which of the following bacterial groups grows best in low oxygen concentrations?
- Obligate aerobes
 - Obligate anaerobes
 - Facultative anaerobes
 - ☒ Microaerophilic bacteria
79. What is the primary temperature for the growth of most pathogenic bacteria?
- 0–4 °C
 - 20–25 °C
 - ☒ 37 °C
 - 42 °C